WHITE PAPER

Breaking Down the Agent2Agent (A2A) Protocol

The New Open Standard for AI Agent Interoperability

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Executive Summary

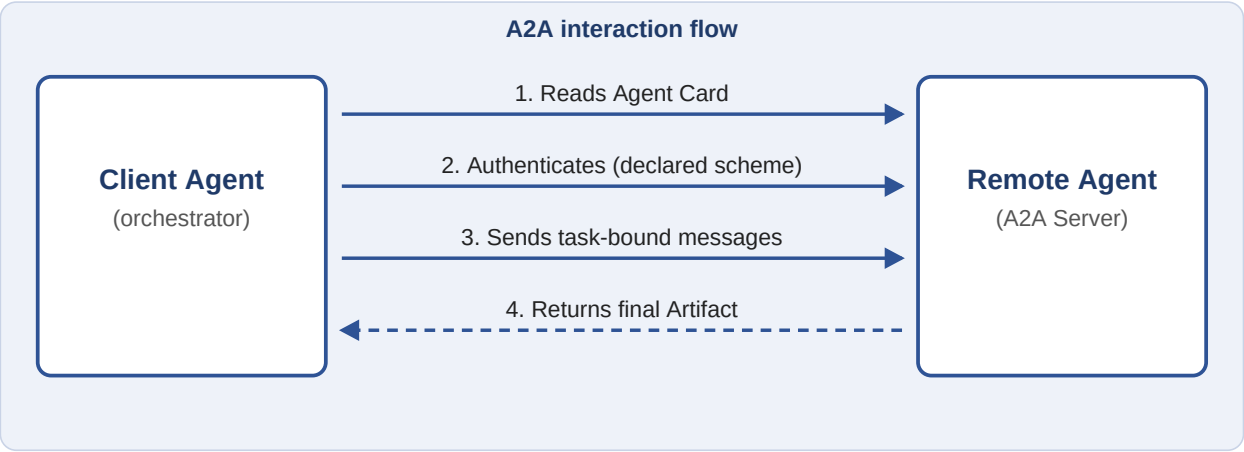
As organizations rapidly deploy autonomous AI agents, a critical bottleneck has emerged: siloed capabilities. A support agent, a pricing agent, and a system health monitor all speak different languages, leading to rigid, brittle point-to-point integrations.

The Agent2Agent (A2A) Protocol is an open, vendor-neutral standard designed to solve this problem. By establishing a universal communication layer, A2A allows specialized AI agents across different platforms to securely discover one another, exchange structured context, and delegate tasks. Initiated by Google in April 2025 and now hosted by the Linux Foundation, A2A moves multi-agent collaboration from custom-coded integrations to a scalable enterprise architecture. The specification has since reached version 1.0.

The Architecture: How A2A Works

The protocol operates on a structured client-server model tailored for software reasoning. Roles are fluid: any agent can act as the client or the server depending on the interaction.

- **The Client Agent** acts as the orchestrator (an application, human assistant, or master agent). It initiates requests and delegates work based on dynamic capabilities rather than hard-coded paths.
- **The Remote Agent (A2A Server)** receives the request, processes the task, and returns the result. Crucially, its internal reasoning and models remain opaque, protecting proprietary logic.



Core Communication Components

To ensure predictability, the protocol standardizes four structural elements:

Element	Role in the protocol
Agent Card	A machine-readable “digital résumé” published at a well-known URL. It lists the agent’s skills, endpoint, input/output modes, and the authentication schemes it accepts.
Task	The central unit of work, assigned a unique ID so progress can be tracked statefully across multi-turn exchanges.
Message	The structured transmission carrying instructions or real-time updates within a task.
Artifact	The final structured output (for example a data file or report) returned by the remote agent.

A2A vs. MCP: Understanding the AI Stack

A frequent point of confusion is how A2A compares to the Model Context Protocol (MCP). They are not competitors; they operate at different layers of the AI stack.

Feature	Agent2Agent (A2A)	Model Context Protocol (MCP)
Primary focus	Agent-to-agent collaboration	LLM-to-tool and data access
Communication target	Lateral: between software agents	Outward: from a model to an external API or database
Ultimate goal	Coordinated multi-agent workflows	Standardized context retrieval for LLMs

> ***Analogy:*** *MCP helps a model look outward to pull data; A2A helps agents work sideways to collaborate once that data is in play.*

Enterprise Security & Technical Foundation

A2A does not reinvent the wheel. It is built on proven, enterprise-grade web standards, which makes it easier to deploy within existing infrastructure.

- **Transport and security:** Runs over HTTP(S). Each agent declares its supported authentication schemes (such as OAuth 2.0, API keys, or OpenID Connect) in its Agent Card, and the server authenticates every incoming request against those declared requirements.
- **Data exchange:** Uses JSON-RPC 2.0 for structured messaging and Server-Sent Events (SSE) for asynchronous, real-time status updates. Later revisions of the specification also add a gRPC binding.
- **Traceability:** Because every interaction is tied to a unique Task ID, enterprises get a clear trail of how agents pass data and hand off work, which supports auditing and observability.

Real-World Business Impact

1. Customer Experience

A front-end triage agent receives a complex customer issue. Using A2A, it queries a backend knowledge agent for technical documentation and a system status agent for active outages. The triage agent unifies these outputs into a single, comprehensive summary for the human representative.

2. Revenue Operations and Sales

A sales agent generates a custom deal quote. Recognizing a heavy discount, it triggers an A2A task to a pricing agent to evaluate dynamic margin limits. If thresholds are crossed, a compliance agent is brought into the loop to validate terms, with all context preserved.

3. Supply Chain Logistics

An inventory monitoring agent flags a low-stock event. It passes a task to a purchasing agent, which routes an order request to an external supplier's agent. The shipping manifest returns as a traceable artifact, closing the supply loop without manual handoffs.

Looking Forward: Interoperability as Infrastructure

As multi-agent ecosystems mature, corporate agility will be determined by how quickly agents can coordinate across different cloud environments and software vendors.

Because A2A is an open infrastructure standard rather than a proprietary vendor feature, it reduces the risk of ecosystem lock-in. It lays the groundwork for a future where AI specialization is expected, and secure, lateral collaboration is built into the foundation.

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Sources: A2A Project specification (a2a-protocol.org), hosted by the Linux Foundation.
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